Power BI Build Guide — Census 2011 Educational Attainment Report

I can't generate a `.pbix` file directly — it's a proprietary binary Power BI Desktop has to write, and I don't have the app or network access here. What I've done instead:

1. **Reshaped the data into a proper star-schema fact table** (`Fact_Education_Long.csv`) so it drops into Power BI cleanly, with real slicers instead of 39 wide columns.
2. **Written the exact DAX measures and page layout below** so you can build the real report in Power BI Desktop in about 15 minutes.
3. **Built a working interactive preview** (the `CensusDashboard` artifact) so you can see the intended report — layout, visuals, cross-filtering — right now, before opening Power BI at all.

1. Import the data

Use **`Fact_Education_Long.csv`** (122,148 rows), not the wide `C08_clean.csv`. It's already unpivoted:

Column	Type	Notes
`State_Code`	Whole number	
`State`	Text	36 values incl. `INDIA`
`Area_Type`	Text	`Total` / `Rural` / `Urban`
`Age_Group`	Text	`'0-6'`, `7` … `19`, `'20-24'` … `'80+'`, `'All ages'`, `'Age not stated'`
`Education_Level`	Text	`Total`, `Illiterate`, `Literate`, or one of 10 attainment levels
`Gender`	Text	`Persons` / `Males` / `Females`
`Population`	Whole	the one measure column

	number	

Power BI → **Get Data → Text/CSV** → point at `Fact_Education_Long.csv` → Load.

Important filter to set on load (Power Query): the `Education_Level` column mixes aggregates (`Total` , `Illiterate` , `Literate`) with the 10 breakdown categories that sum up to `Literate` . Most visuals should filter to one or the other — don't sum across both or you'll double-count. I recommend adding two calculated columns in Power Query:

- `Level_Type` **=** `IF Education_Level IN {"Total","Illiterate","Literate"} THEN "Aggregate" ELSE "Attainment"`

2. Key DAX measures

```
Total Population =
CALCULATE(SUM(Fact_Education_Long[Population]), Fact_Education_Long[Education_Lev

Total Population (7+) =
CALCULATE([Total Population], Fact_Education_Long[Age_Group] <> "0-6")

Literate Population =
CALCULATE(SUM(Fact_Education_Long[Population]), Fact_Education_Long[Education_Lev

Illiterate Population =
CALCULATE(SUM(Fact_Education_Long[Population]), Fact_Education_Long[Education_Lev

Literacy Rate % =
DIVIDE([Literate Population], [Total Population (7+)])

-- Male/Female literacy: filter Gender = "Males"/"Females" in the visual or a
-- separate explicit measure:
Male Literacy % =
DIVIDE(
    CALCULATE(SUM(Fact_Education_Long[Population]),
        Fact_Education_Long[Education_Level] = "Literate", Fact_Education_Long[Ge
    CALCULATE(SUM(Fact_Education_Long[Population]),
        Fact_Education_Long[Education_Level] = "Total", Fact_Education_Long[Gende
        Fact_Education_Long[Age_Group] <> "0-6")
)

Female Literacy % =
DIVIDE(
    CALCULATE(SUM(Fact_Education_Long[Population]),
        Fact_Education_Long[Education_Level] = "Literate", Fact_Education_Long[Ge
    CALCULATE(SUM(Fact_Education_Long[Population]),
```

```
        Fact_Education_Long[Education_Level] = "Total", Fact_Education_Long[Gende
        Fact_Education_Long[Age_Group] <> "0-6")
)

Graduates & Above =
CALCULATE(SUM(Fact_Education_Long[Population]),
    Fact_Education_Long[Education_Level] = "Graduate & Above", Fact_Education_Lon
```

> **Why exclude age 0-6 from the literacy-rate denominator?** The Census itself scopes this table to "population age 7 and above," but the `'All ages'` aggregate row still includes the 0-6 population (counted as illiterate by definition). Using it as-is understates literacy by ~10 points vs. the officially published rate. All measures above match the official ~74% national literacy figure.

3. Report pages (mirrors the interactive preview)

Page 1 — Overview

- KPI cards: `Total Population (7+)`, `Literacy Rate %`, `Illiterate Population`, `Graduates & Above`
- Horizontal bar chart: `Literacy Rate %` by `State` (filter out `INDIA`), sorted descending
- Donut chart: `Population` by `Education_Level` (filter `Level_Type = "Attainment"`, `Gender = "Persons"`)
- Slicers: `Area_Type` (Total/Rural/Urban button slicer), `State` (dropdown)

Page 2 — Gender Gap

- Clustered bar: `Male Literacy %` and `Female Literacy %` by `State`, sorted by gap
- Card: national `Male Literacy %` vs `Female Literacy %`

Page 3 — Rural vs Urban

- Clustered bar: `Literacy Rate %` by `State`, split by `Area_Type` (Rural/Urban only)

Page 4 — Age & Attainment

- Line/area combo chart: `Illiterate Population` and `Graduates & Above`, both as % of `Total Population`, by `Age_Group` (exclude `All ages`, `Age not stated`, `0-6`), Area = India, Total only
- Sort `Age_Group` using a custom sort column (numeric order, not alphabetical — `'20-`

24' should sort after `19`, not before `2`)

Page 5 — Education Mix

- 100%-stacked bar: `Population` by `State`, stacked by `Education_Level` (`Level_Type = "Attainment"`, `Gender = "Persons"`)

4. Cross-filtering behavior

Set every visual's **Edit interactions** to cross-filter (default), so clicking a state bar on Page 1 filters the KPI cards and other visuals on that page — this is what the interactive preview simulates in the browser.

Files referenced

- `Fact_Education_Long.csv` — import this into Power BI
- `C08_clean.csv` / `C08_clean.xlsx` — wide format, kept for reference/Excel use